

Ying Zhao

👤 Profile

I am a passionate PhD researcher in materials science with expertise in atomistic and mesoscale simulations, including density functional theory and molecular dynamics. My work focuses on thermodynamics, kinetics, interfaces, and defects in low-dimensional materials for energy and electronic applications. I am committed to advancing sustainable technologies through physics-based modeling and quantitative simulations, translating fundamental insights into practical solutions for energy systems, sensors, and next-generation electronics.

🎓 Education

PhD in Materials Science, McGill University, Montréal

September 2021 — Present

- **Research Group:** McGill NanoFactory
- **Research Area:** Atomistic and mesoscale simulations of low-dimensional materials for energy and electronic applications
- **Awards:** McGill Engineering Doctoral Award, McGill Graduate Excellence Award

Master of Materials Science, McGill University, Montréal

January 2020 — December 2021

- **Research Topic:** Grain boundary engineering in 2D TMDCs for electrocatalytic CO₂ reductions
- **Award:** McGill Graduate Excellence Award

Bachelor of Materials Engineering, McGill University, Montréal

September 2014 — April 2019

- **Award:** MTL Women's Association of the Canadian Institute of Mining, Metallurgy, and Petroleum Bursary

💼 Employment History

Research Assistance at McGill University, Montréal

September 2019 — December 2019

- Conducted academic research utilizing atomistic simulations and theoretical tools to investigate low dimensional nanomaterials with topological features for photovoltaic and electrocatalytic applications

Environmental Engineer (Co-op) at Nexen Energy ULC, Calgary

May 2017 — August 2018

- Collected operating data, performed gas emissions calculations, and generated emission reports to meet federal and provincial requirements, for example, annual national pollutant release inventory (NPRI) report, greenhouse gas report, etc.
- Created an Excel-based automated compliance calendar (with macros), streamlining tracking of all past and upcoming environmental obligations, which reduced non-compliance incidence to zero

Contact

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LinkedIn | **ResearchGate** | **Google Scholar**

Skills

Materials Simulation:

Ab initio density functional theory (VASP), molecular dynamics (LAMMPS), thermodynamics (FactSage), fluid dynamics (ANSYS)

Microsoft Office Suite:

PowerPoint, Excel, Word

Programming:

Python (NumPy, Pandas, Scikit-learn, PyTorch, TensorFlow, Pymatgen, Atomate, Fireworks), MATLAB, C, Shell Scripting

Data Management:

MongoDB

Machine Learning:

Model development and optimization, feature engineering, deep learning (PyTorch, TensorFlow), data visualization, predictive analytics

Version Control:

Git, GitHub

Materials Analyzing and Manufacturing:

Flexural, buckling, torsion, hardness, and tensile testing; hot and cold rolling; XRD; SEM; XPS;

Selected Publications

Activation of Electrocatalytic CO₂ Methanation on Defective 1T' Phase TMDCs, *ACS Appl. Energy Mater.*

Zhao, Y.; Chen, Y.; Ou, P.; Song, J., January 2026

- Conducted DFT-based atomistic simulations to investigate CO₂ electroreduction on semimetallic 1T' phase TMDCs (MoTe₂, WTe₂), focusing on basal-plane activity and defect effects.
- Applied random forest machine learning models to correlate electronic and structural descriptors (e.g., metal–oxygen bond lengths, substrate deformation, etc.) with catalytic activity, guiding defect engineering strategies.
- Presented at 2023 Canadian Materials Science Conference (CMSC)

Efficient Hole Extraction and *OH Alleviation by Pd Nanoparticles on GaN Nanowires in Seawater for Solar-Driven H₂ and H₂O₂ Generation, *Angewandte Chemie Int. Ed.*

Salman Nasir*, M.; Zhao, Y.*; Ye, H.; Wang, T.; Sheng, B.; Song, J.; Li, J.; Wang, P.; Wang, X.; Huang, Z.; Zhou, B., March 2025

- Collaborated on the project integrating Pd nanoparticles with GaN nanowires (Pd NPs/GaN NWs) for photocatalytic seawater splitting to simultaneously produce H₂ and H₂O₂ under sunlight.
- Performed DFT calculations to validate experimental results, revealing that Pd NPs act as efficient hole extractors and *OH alleviators, and that chloride ions facilitate the H₂O₂ formation pathway by lowering energy barriers.

Van der Waals quantum dots on layered hexagonal boron nitride, *PNAS*

Wu, Y.*; Xiao, Y.*; Zhao, Y.*; Shen, Y.; Sun, K.; Wang, B.; Wang, P.; Wang, D.; Zhou, P.; Wang, D.; Liu, J.; Hu, M.; B. Norris, T.; Song, J.; Mi, Z., February 2025

- Collaborated on a multidisciplinary project to synthesize GaN van der Waals quantum dots (vQDs) on hexagonal boron nitride (hBN) without wetting layer formation, enabling mechanical exfoliation from epitaxial substrates.
- Conducted DFT and ab-initio molecular dynamics (AIMD) calculations to explain experimental observations, showing that interadatom bond strengths are an order of magnitude stronger than bonds to the hBN substrate, providing a theoretical basis for vQD stability and optical performance.
- Proposed a general formation mechanism for vQDs, highlighting their potential for high-performance optoelectronic and quantum devices.

Basal plane activation of two-dimensional transition metal dichalcogenides via alloying for hydrogen evolution reaction: first-principles calculations and machine learning prediction, *J. Mater. Chem. A*

Chen, Y.; Zhao, Y.; Ou, P.; Song, J., January 2023

- Assisted developing a machine learning workflow based on first-principles calculations to predict the HER activity and stability for a series of 2D cation-mixed TMDC alloys of various compositions

Basal Plane Activation via Grain Boundaries in Monolayer MoS₂ for Carbon Dioxide Reduction, *ACS Catal.*

Zhao, Y.; Chen, Y.; Ou, P.; Song, J., September 2023

- Performed DFT calculations to study CO₂ electroreduction on polycrystalline MoS₂, focusing on the role of grain boundaries (GBs) and sulfur vacancies.
- Presented at 2023 Canadian Materials Science Conference (CMSC)

*Co-first authors

AAS; fractography; powder metallurgy

Language:

Fluent in English and Mandarin;
Basic comprehension of French

Design Tools:

AutoCAD, 3ds Max, Adobe Illustrator

Critical Thinking & Problem-Solving

Collaboration & Teamwork

Attention to Detail

Adaptability & Continuous Learning

Certifications

Workplace Hazardous Materials Information System (WHMIS), Montréal

September 2018, McGill University

3D Modelling and Animation, Calgary

December 2017, Southern Alberta Institute of Technology (SAIT)